

Spring Assessment

Section A – Theory Answers

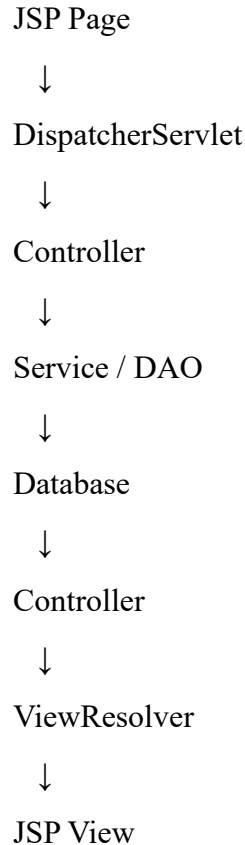
1) Explain the flow of a request in Spring MVC

Answer:

❖ **In Spring MVC, the request follows these steps:**

- The user submits data from a JSP form.
- The request goes to the DispatcherServlet.
- DispatcherServlet finds the appropriate Controller.
- The Controller processes the request.
- DAO or Service layer interacts with the database.
- Data is returned to the Controller.
- The Controller sends the model data and view name.
- ViewResolver maps the logical view name to a JSP page.
- The final response is displayed to the user.

➤ **Flow Diagram**



2) Explain IoC and Dependency Injection

Answer:

❖ IoC (Inversion of Control)

- IoC means the control of object creation is given to the Spring Container instead of the programmer.
- **Example without Spring:**

```
Student s = new Student();
```
- In Spring, the container creates and manages the object automatically.

❖ Dependency Injection

- Dependency Injection means providing required objects to a class by the Spring framework.
- **Types of Dependency Injection**
 1. Constructor Injection
 2. Setter Injection
- **Example**

```
@Autowired  
private UserDao dao;
```
- Spring automatically injects the object.

3) Differentiate between BeanFactory and ApplicationContext

Answer:

BeanFactory	ApplicationContext
Basic container	Advanced container
Lazy initialization	Eager initialization
Less features	More features
No event support	Supports events
Used in small applications	Mostly used in enterprise applications

4) Explain how JdbcTemplate simplifies JDBC operations

Answer:

JdbcTemplate reduces boilerplate JDBC code.

➤ Advantages

- Automatic connection management
- Automatic exception handling
- Automatically closes resources
- Less coding required

- **Example**

```
String sql = "select * from user";  
List<User> users = jdbcTemplate.query(sql, new  
BeanPropertyRowMapper<>(User.class));
```

5) Explain the role of DispatcherServlet in Spring MVC

Answer:

DispatcherServlet is the front controller of Spring MVC.

➤ **Responsibilities**

- Receives all requests
- Finds appropriate controller
- Sends request to controller
- Handles response
- Resolves views using ViewResolver

6) Explain constructor injection vs setter injection

Answer:

Constructor Injection	Setter Injection
Dependency injected through constructor	Dependency injected through setter method
Used for mandatory dependency	Used for optional dependency
Object becomes immutable	Object can be modified

- **Constructor Injection Example**

```
public Student(Service s){  
    this.s = s;  
}
```

- **Setter Injection Example**

```
public void setService(Service s){  
    this.s = s;  
}
```
